



NUST CHIP DESIGN CENTRE

RISC-V Architecture

Lab Manual # 26

Cache Design in SystemVerilog

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1. In-Lab Tasks:

i. Task 1:

```
Line 4060 differs:
../log/access.log: MISS! ; PA: 5327 ; Accessed Data: 0
../log/access.log: MISS! ; PA: 5327 ; Accessed Data: 0

Line 4066 differs:
../log/access.log: HIT! ; PA: 9003 ; Accessed Data: 11011010010011110100011110111110
../log/access.log: HIT! ; PA: 9003 ; Accessed Data: 11011010010011110100011110111110

Line 4069 differs:
../log/access.log: MISS! ; PA: 9033 ; Accessed Data: 0
../log/access.log: MISS! ; PA: 9033 ; Accessed Data: 0

Line 4083 differs:
../log/access.log: MISS! ; PA: 6414 ; Accessed Data: 0
../log/access.log: MISS! ; PA: 6414 ; Accessed Data: 0

Line 4089 differs:
../log/access.log: MISS! ; PA: 8541 ; Accessed Data: 0
../log/access.log: MISS! ; PA: 8541 ; Accessed Data: 0

Line 4093 differs:
../log/access.log: MISS! ; PA: 2454 ; Accessed Data: 0
../log/access.log: MISS! ; PA: 2454 ; Accessed Data: 0

Tests Completed
Differences -> 656
PS F:\muddassar_ali_siddiqui\ncdc\riscv\architecture\labs\lab26\tester_files\cache-tester\scripts>
```

The difference is just padding. It is the python script difference it generates a file without padding while our given testbench gives with padding. In task 2 and 3 I'll resolve the issue.

ii. Task 2:

```
PS F:\muddassar_ali_siddiqui\ncdc\riscv\architecture\labs\lab26\tester_files\cache-tester\scripts> python ./cache_init.py -b 128 -w 4 -a 2
Physical Address Length: 16 ; Tag Length: 5 ;
Updates in Way 0:
Updated 18 blocks
Updates in Way 1:
Updated 17 blocks
Replaced 35 35 126 1
Cache Size (Ways,Fields,Blocks)(2, 6, 128)
PS F:\muddassar_ali_siddiqui\ncdc\riscv\architecture\labs\lab26\tester_files\cache-tester\scripts> python ./check.py ../log/access.log ../log/core_access.log

Tests Completed
Differences -> 0
PS F:\muddassar_ali_siddiqui\ncdc\riscv\architecture\labs\lab26\tester_files\cache-tester\scripts>
```

The files are identical and there is no difference.

iii. Task 3:

```
PS F:\muddassir_ali_siddiqui\ncdc\riscv\architecture\labs\lab26\tester_files\cache-tester\scripts> python ./cache_init.py -b 1 -w 4 -a 256  
Physical Address Length: 16 ; Tag Length: 12 ;
```

Updates in Way 27: Updated 1 blocks	Updates in Way 53: Updated 0 blocks	Updated 1 blocks Updates in Way 80: Updated 1 blocks	Updates in Way 106: Updated 1 blocks	Updates in Way 0: Updated 1 blocks
Updates in Way 28: Updated 1 blocks	Updates in Way 54: Updated 0 blocks	Updated 1 blocks Updates in Way 81: Updated 0 blocks	Updates in Way 107: Updated 1 blocks	Updates in Way 1: Updated 0 blocks
Updates in Way 29: Updated 0 blocks	Updates in Way 55: Updated 1 blocks	Updated 0 blocks Updates in Way 82: Updated 0 blocks	Updates in Way 108: Updated 1 blocks	Updates in Way 2: Updated 0 blocks
Updates in Way 30: Updated 0 blocks	Updates in Way 56: Updated 1 blocks	Updated 0 blocks Updates in Way 83: Updated 1 blocks	Updates in Way 109: Updated 1 blocks	Updates in Way 3: Updated 1 blocks
Updates in Way 31: Updated 1 blocks	Updates in Way 57: Updated 0 blocks	Updated 1 blocks Updates in Way 84: Updated 1 blocks	Updates in Way 110: Updated 1 blocks	Updates in Way 4: Updated 1 blocks
Updates in Way 32: Updated 1 blocks	Updates in Way 58: Updated 0 blocks	Updated 1 blocks Updates in Way 85: Updated 0 blocks	Updates in Way 111: Updated 1 blocks	Updates in Way 5: Updated 0 blocks
Updates in Way 33: Updated 0 blocks	Updates in Way 59: Updated 1 blocks	Updated 0 blocks Updates in Way 86: Updated 0 blocks	Updates in Way 112: Updated 1 blocks	Updates in Way 6: Updated 0 blocks
Updates in Way 34: Updated 0 blocks	Updates in Way 60: Updated 1 blocks	Updated 0 blocks Updates in Way 87: Updated 1 blocks	Updates in Way 113: Updated 1 blocks	Updates in Way 7: Updated 1 blocks
Updates in Way 35: Updated 1 blocks	Updates in Way 61: Updated 0 blocks	Updated 1 blocks Updates in Way 88: Updated 1 blocks	Updates in Way 114: Updated 1 blocks	Updates in Way 8: Updated 1 blocks
Updates in Way 36: Updated 1 blocks	Updates in Way 62: Updated 0 blocks	Updated 1 blocks Updates in Way 89: Updated 0 blocks	Updates in Way 115: Updated 1 blocks	Updates in Way 9: Updated 0 blocks
Updates in Way 37: Updated 0 blocks	Updates in Way 63: Updated 1 blocks	Updated 0 blocks Updates in Way 90: Updated 0 blocks	Updates in Way 116: Updated 1 blocks	Updates in Way 10: Updated 0 blocks
Updates in Way 38: Updated 0 blocks	Updates in Way 64: Updated 1 blocks	Updated 0 blocks Updates in Way 91: Updated 1 blocks	Updates in Way 117: Updated 1 blocks	Updates in Way 11: Updated 1 blocks
Updates in Way 39: Updated 1 blocks	Updates in Way 65: Updated 0 blocks	Updated 1 blocks Updates in Way 92: Updated 1 blocks	Updates in Way 118: Updated 1 blocks	Updates in Way 12: Updated 1 blocks
Updates in Way 40: Updated 1 blocks	Updates in Way 66: Updated 0 blocks	Updated 1 blocks Updates in Way 93: Updated 0 blocks	Updates in Way 119: Updated 1 blocks	Updates in Way 13: Updated 0 blocks
Updates in Way 41: Updated 0 blocks	Updates in Way 67: Updated 1 blocks	Updated 0 blocks Updates in Way 94: Updated 0 blocks	Updates in Way 120: Updated 1 blocks	Updates in Way 14: Updated 0 blocks
Updates in Way 42: Updated 0 blocks	Updates in Way 68: Updated 1 blocks	Updated 0 blocks Updates in Way 95: Updated 1 blocks	Updates in Way 121: Updated 1 blocks	Updates in Way 15: Updated 1 blocks
Updates in Way 43: Updated 1 blocks	Updates in Way 69: Updated 0 blocks	Updated 1 blocks Updates in Way 96: Updated 1 blocks	Updates in Way 122: Updated 1 blocks	Updates in Way 16: Updated 1 blocks
Updates in Way 44: Updated 1 blocks	Updates in Way 70: Updated 0 blocks	Updated 1 blocks Updates in Way 97: Updated 0 blocks	Updates in Way 123: Updated 1 blocks	Updates in Way 17: Updated 0 blocks
Updates in Way 45: Updated 0 blocks	Updates in Way 71: Updated 1 blocks	Updated 0 blocks Updates in Way 98: Updated 0 blocks	Updates in Way 124: Updated 1 blocks	Updates in Way 18: Updated 0 blocks
Updates in Way 46: Updated 0 blocks	Updates in Way 72: Updated 1 blocks	Updated 0 blocks Updates in Way 99: Updated 1 blocks	Updates in Way 125: Updated 1 blocks	Updates in Way 19: Updated 1 blocks
Updates in Way 47: Updated 1 blocks	Updates in Way 73: Updated 0 blocks	Updated 1 blocks Updates in Way 100: Updated 1 blocks	Updates in Way 126: Updated 1 blocks	Updates in Way 20: Updated 1 blocks
Updates in Way 48: Updated 1 blocks	Updates in Way 74: Updated 0 blocks	Updated 1 blocks Updates in Way 101: Updated 0 blocks	Updates in Way 127: Updated 1 blocks	Updates in Way 21: Updated 0 blocks
Updates in Way 49: Updated 0 blocks	Updates in Way 75: Updated 1 blocks	Updated 0 blocks Updates in Way 102: Updated 0 blocks	Updates in Way 128: Updated 1 blocks	Updates in Way 22: Updated 0 blocks
Updates in Way 50: Updated 0 blocks	Updates in Way 76: Updated 1 blocks	Updated 0 blocks Updates in Way 103: Updated 1 blocks	Updates in Way 129: Updated 1 blocks	Updates in Way 23: Updated 1 blocks
Updates in Way 51: Updated 1 blocks	Updates in Way 77: Updated 0 blocks	Updated 1 blocks Updates in Way 104: Updated 1 blocks	Updates in Way 130: Updated 1 blocks	Updates in Way 24: Updated 1 blocks
Updates in Way 52: Updated 0 blocks	Updates in Way 78: Updated 0 blocks	Updated 1 blocks Updates in Way 105: Updated 0 blocks	Updates in Way 131: Updated 1 blocks	Updates in Way 25: Updated 0 blocks
Updated 1 blocks	Updates in Way 79: Updated 0 blocks	Updated 0 blocks	Updates in Way 132: Updated 0 blocks	Updated 0 blocks

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Updated 1 blocks	Updates in Way 159:	Updated 1 blocks	Updates in Way 212:
Updated in Way 133:	Updated 0 blocks	Updated in Way 186:	Updated 0 blocks
Updated 1 blocks	Updates in Way 160:	Updated 1 blocks	Updates in Way 213:
Updated in Way 134:	Updated 0 blocks	Updated in Way 187:	Updated 1 blocks
Updated 1 blocks	Updates in Way 161:	Updated 0 blocks	Updates in Way 214:
Updated in Way 135:	Updated 1 blocks	Updated in Way 188:	Updated 1 blocks
Updated 1 blocks	Updates in Way 162:	Updated 0 blocks	Updates in Way 215:
Updated in Way 136:	Updated 1 blocks	Updated in Way 189:	Updated 0 blocks
Updated 1 blocks	Updates in Way 163:	Updated 1 blocks	Updates in Way 216:
Updated in Way 137:	Updated 0 blocks	Updated in Way 190:	Updated 0 blocks
Updated 1 blocks	Updates in Way 164:	Updated 1 blocks	Updates in Way 217:
Updated in Way 138:	Updated 0 blocks	Updated in Way 191:	Updated 1 blocks
Updated 1 blocks	Updates in Way 165:	Updated 0 blocks	Updates in Way 218:
Updated in Way 139:	Updated 1 blocks	Updated in Way 192:	Updated 1 blocks
Updated 1 blocks	Updates in Way 166:	Updated 0 blocks	Updates in Way 219:
Updated in Way 140:	Updated 1 blocks	Updated in Way 193:	Updated 0 blocks
Updated 1 blocks	Updates in Way 167:	Updated 1 blocks	Updates in Way 220:
Updated in Way 141:	Updated 0 blocks	Updated in Way 194:	Updated 0 blocks
Updated 1 blocks	Updates in Way 168:	Updated 1 blocks	Updates in Way 221:
Updated in Way 142:	Updated 0 blocks	Updated in Way 195:	Updated 1 blocks
Updated 1 blocks	Updates in Way 169:	Updated 0 blocks	Updates in Way 222:
Updated in Way 143:	Updated 1 blocks	Updated in Way 196:	Updated 1 blocks
Updated 1 blocks	Updates in Way 170:	Updated 0 blocks	Updates in Way 223:
Updated in Way 144:	Updated 1 blocks	Updated in Way 197:	Updated 0 blocks
Updated 1 blocks	Updates in Way 171:	Updated 1 blocks	Updates in Way 224:
Updated in Way 145:	Updated 0 blocks	Updated in Way 198:	Updated 0 blocks
Updated 1 blocks	Updates in Way 172:	Updated 1 blocks	Updates in Way 225:
Updated in Way 146:	Updated 0 blocks	Updated in Way 199:	Updated 1 blocks
Updated 1 blocks	Updates in Way 173:	Updated 0 blocks	Updates in Way 226:
Updated in Way 147:	Updated 1 blocks	Updated in Way 200:	Updated 1 blocks
Updated 1 blocks	Updates in Way 174:	Updated 0 blocks	Updates in Way 227:
Updated in Way 148:	Updated 1 blocks	Updated in Way 201:	Updated 0 blocks
Updated 1 blocks	Updates in Way 175:	Updated 1 blocks	Updates in Way 228:
Updated in Way 149:	Updated 0 blocks	Updated in Way 202:	Updated 0 blocks
Updated 1 blocks	Updates in Way 176:	Updated 1 blocks	Updates in Way 229:
Updated in Way 150:	Updated 0 blocks	Updated in Way 203:	Updated 1 blocks
Updated 1 blocks	Updates in Way 177:	Updated 0 blocks	Updates in Way 230:
Updated in Way 151:	Updated 1 blocks	Updated in Way 204:	Updated 1 blocks
Updated 0 blocks	Updates in Way 178:	Updated 0 blocks	Updates in Way 231:
Updated in Way 152:	Updated 1 blocks	Updated in Way 205:	Updated 0 blocks
Updated 0 blocks	Updates in Way 179:	Updated 1 blocks	Updates in Way 232:
Updated in Way 153:	Updated 0 blocks	Updated in Way 206:	Updated 0 blocks
Updated 1 blocks	Updates in Way 180:	Updated 1 blocks	Updates in Way 233:
Updated in Way 154:	Updated 0 blocks	Updated in Way 207:	Updated 1 blocks
Updated 1 blocks	Updates in Way 181:	Updated 0 blocks	Updates in Way 234:
Updated in Way 155:	Updated 1 blocks	Updated in Way 208:	Updated 1 blocks
Updated 0 blocks	Updates in Way 182:	Updated 0 blocks	Updates in Way 235:
Updated in Way 156:	Updated 1 blocks	Updated in Way 209:	Updated 0 blocks
Updated 0 blocks	Updates in Way 183:	Updated 1 blocks	Updates in Way 236:
Updated in Way 157:	Updated 0 blocks	Updated in Way 210:	Updated 0 blocks
Updated 1 blocks	Updates in Way 184:	Updated 1 blocks	Updates in Way 237:
Updated in Way 158:	Updated 0 blocks	Updated in Way 211:	Updated 1 blocks
Updated 1 blocks	Updates in Way 185:	Updated 0 blocks	Updates in Way 238:

Updated 1 blocks
Updates in Way 239:
Updated 0 blocks
Updates in Way 240:
Updated 0 blocks
Updates in Way 241:
Updated 1 blocks
Updates in Way 242:
Updated 1 blocks
Updates in Way 243:
Updated 0 blocks
Updates in Way 244:
Updated 0 blocks
Updates in Way 245:
Updated 1 blocks
Updates in Way 246:
Updated 1 blocks
Updates in Way 247:
Updates in Way 248:
Updates in Way 249:
Updated 1 blocks
Updates in Way 250:
Updated 1 blocks
Updates in Way 251:
Updated 0 blocks
Updates in Way 252:
Updated 0 blocks
Updates in Way 253:
Updated 1 blocks
Updates in Way 254:
Updated 1 blocks
Updates in Way 255:
Updated 0 blocks

```
Replaced 150 150 0 0
Cache Size (Ways,Fields,Blocks)(256, 6, 1)
PS F:\muddassir_ali_siddiqui\ncdc\riscv\architecture\labs\lab26\tester_files\cache-tester\scripts> python ./check.py ../log/access.log ../log/core_access.log
Files have different number of lines.
PS F:\muddassir_ali_siddiqui\ncdc\riscv\architecture\labs\lab26\tester_files\cache-tester\scripts> python ./check.py ../log/access.log ../log/core_access.log
Tests Completed
Differences -> 0
PS F:\muddassir_ali_siddiqui\ncdc\riscv\architecture\labs\lab26\tester_files\cache-tester\scripts> |
```

The files are identical and has no difference.

Nust Chip Design Center (NCDC)

2. Critical Analysis:

In Lab 26 we designed and verified three read-only cache architectures in SystemVerilog: direct-mapped, two-way set-associative, and fully associative. Each cache stores four 32-bit words per block and was initialized from the provided binary files, so no write or replacement logic was required. The direct-mapped cache used a single tag match per access, the two-way set-associative design compared tags in two parallel ways for each set, and the fully associative cache checked all tags for every read. We focused on correct address breakdown into tag, index, word, and byte offsets to ensure accurate hit detection. Simulation with the supplied testbench and logs confirmed functional operation and highlighted how increasing associativity improves flexibility and potential hit rate while adding hardware complexity.